QUASI-COHERENT MODULES ON A CHAOTIC SITE WITH FLAT RESTRICTIONS FORM A WEAK SERRE SUBCATEGORY

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains a commutative diagram typeset with Xy-pic; the retained two-slice preamble of this source supplies the native Xy-pic setup, so it compiles as in the source.

Editorial note. The counted span cites two external items (an example and a definition), recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Characterization of quasi-coherence in the chaotic topology by base change. Stacks Project, Modules on Sites, Lemma lemma-chaotic-quasi-coherent (tag 09BA). Turns quasi-coherence into the module-theoretic base-change condition that the diagram chase tests.
- Weak Serre subcategories and the five lemma. Stacks Project, Homology, Definition homology-definition-serre-subcategory (tag 02MP). Supplies the property to be verified and the tool for the middle-term conclusion.

Source reference. Original reference 07GE: sources/stacks-chow/tags.txt, source lines 9514–9514. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02MO: sources/stacks-chow/tags.txt, source lines 3451–3451. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GZN: sources/stacks-chow/tags.txt, source lines 21183–21183. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 05QB: sources/stacks-chow/tags.txt, source lines 7370–7370. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{C}$ be a category viewed as a site with the chaotic topology, see Sites, Example 07GE. Let $\mathcal{O}$ be a sheaf of rings on $\mathcal{C}$. Assume for all $U \to V$ in $\mathcal{C}$ the restriction map $\mathcal{O}(V) \to \mathcal{O}(U)$ is a flat ring map. Then the category of quasi-coherent $\mathcal{O}$-modules is a weak Serre subcategory of $Mod(\mathcal{O})$.*

Proof. We will check the definition of a weak Serre subcategory, see Homology, Definition 02MO. To do this we will use the characterization of quasi-coherent modules given in Lemma 0GZN. Consider an exact sequence

$$\mathcal{F}_0 \to \mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to \mathcal{F}_4$$

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in $Mod(\mathcal{O})$ with $\mathcal{F}_0$, $\mathcal{F}_1$, $\mathcal{F}_3$, and $\mathcal{F}_4$ quasi-coherent. Let $U \to V$ be a morphism of $\mathcal{C}$ and consider the commutative diagram

$$\begin{array}{ccccccccc}
\mathcal{F}_0(V) \otimes_{\mathcal{O}(V)} \mathcal{O}(U) & \longrightarrow & \mathcal{F}_1(V) \otimes_{\mathcal{O}(V)} \mathcal{O}(U) & \longrightarrow & \mathcal{F}_2(V) \otimes_{\mathcal{O}(V)} \mathcal{O}(U) & \longrightarrow & \mathcal{F}_3(V) \otimes_{\mathcal{O}(V)} \mathcal{O}(U) & \longrightarrow & \mathcal{F}_4(V \\
\downarrow & & \downarrow & & \downarrow & & \downarrow & & \\
\mathcal{F}_0(U) & \longrightarrow & \mathcal{F}_1(U) & \longrightarrow & \mathcal{F}_2(U) & \longrightarrow & \mathcal{F}_3(U) & \longrightarrow &
\end{array}$$

By assumption the vertical arrows with indices 0, 1, 3, 4 are isomorphisms. Since the topology on $\mathcal{C}$ is chaotic taking sections over an object of $\mathcal{C}$ is exact and hence the lower row is exact. Since $\mathcal{O}(V) \to \mathcal{O}(U)$ is flat also the upper row is exact. Thus we conclude that the middle arrow is an isomorphism by the 5 lemma (Homology, Lemma 05QB). $\square$